Identities used: binomial therom
$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$
pascals identity
$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

Reucrsions from Binomial therom

Non falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$$

let $a = 1$

$$(x+1)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b \right]$$

let $x = x' - 1$

$$((x'-1)+1)^r = x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x'-1)^b \right]$$

in general

$$((x-b)+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-b)^b a^{r-b} \right]$$

if b=a

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b} \right]$$

we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b} \right]$$

$$(x-a)^b = \sum_{c=0}^{b} \left[\binom{b}{c} ((x-a)-a)^c a^{b-c} \right]$$

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} ((x-a)-a)^c a^{b-c} \right] a^{r-b} \right]$$

every layer strips one a from inner power's base

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} ((x-a)-a)^c a^{b-c} \right] a^{r-b} \right]$$

$$x^r = \overbrace{\sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} \sum_{d=0}^{c} \left[\binom{c}{d} \dots \left[\sum_{n_{\frac{x}{a}}=0}^{n_{(\frac{x}{a})-1}} \left[\binom{n_{(\frac{x}{a})-1}}{n_{\frac{x}{a}}} 0^{n_{\frac{x}{a}}} a^{n_{(\frac{x}{a})-1}-n_{\frac{x}{a}}} \right] \right] a^{c-d} \right] a^{b-c} \right] a^{r-b} \right]}^{\frac{x}{a} \text{ sums}}$$

you can stop at any layer you want, you dont need to fully expand

$$x^r = \overbrace{\sum_{b=0}^{r} \left[\binom{r}{b} \sum_{c=0}^{b} \left[\binom{b}{c} \sum_{d=0}^{c} \left[\binom{c}{d} \dots \left[\sum_{n_l=0}^{n_{l-1}} \left[\binom{n_{l-1}}{n_l} (x-la)^{n_l} a^{n_{l-1}-n_l} \right] \right] a^{c-d} \right] a^{b-c} \right] a^{r-b} \right]}^{l \text{ sums}}$$

falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$$

$$(x+a)^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} x^b a^{r-b} \right] + \binom{r}{r} x^r a^0$$

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-a)^b a^{r-b} \right] + \binom{r}{r} (x-a)^r a^0$$

the same reassoning applys here as in the last case:
"we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0 "

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}(x-a)^b\right] + \binom{r}{r}a^0(x-a)^r$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}((x-a)-a)^c\right] + \binom{b}{b}a^0((x-a)-a)^b\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\right.\right.}^{l \text{ sums}}$$
$$\left.\left.\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right] + \binom{n_{l-1}}{n_{l-1}}a^0(x-la)^n_{l-1}\right]\cdots\right.\right.$$
$$\left.\left. + \binom{b}{b}a^0((x-a)-a)^b\right]\right] + \binom{r}{r}(x-a)^r a^0$$

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$$\binom{r}{r} = 1 \text{ for all } r \in \mathbb{Z}^+$$

if we restrict our formula with these we can simplify it

according to our restriciton $a^0 = 1$ and $\binom{r}{r} = 1$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left(\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\left(\cdots\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right] + (x-la)^{n_{l-1}}\cdots\right)\right] + (x-2a)^b\right)\right]}^{l \text{ sums}} + (x-a)^r$$

the sum can be factored into a sum of falling sums The papers theory holds if the restrictions hold:

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$\binom{r}{r} = 1$ for all $r \in \mathbb{Z}^+$ Since we can freely choose a only the restriction on r remains

“I have a theory, all x^n’s can be written as a sum of nested sums with falling upper bounds “

all x^r’s can be written as a sum of nested sums with falling upper bounds for all $r \in \mathbb{Z}^+$ also x must have a divisor d in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right)$, if it has this divisor it can be written a nested sum with l sums for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid d$ in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right) \Rightarrow$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right]\right]\cdots\right]\right]\right]}^{l \text{ sums}}$$
$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\cdots\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}\left[\binom{n_{l-2}}{n_{l-1}}a^{n_{l-2}-n_{l-1}}(x-la)^{n_{l-1}}\right]\right]\cdots\right]\right]\right]}^{l-1 \text{ sums}}$$
$$+\cdots+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[(x-2a)^b\right]\right]}^{1 \text{ sums}}$$
$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

Falling sums (l,r upper bound)

$$(x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

$$(x+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}x^b a^{r-b}\right] + \binom{r}{r}x^r a^0$$

$$\text{let } x' = x-a, \text{let } a' = a$$

$$((x-a)+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

the previous falling sums resulted from expanding the inner (x-a)^b, if we expand the outer (x-a)^r we get a diffrent falling sum

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-a)-a)^b a^{r-b}\right] + \binom{r}{r}((x-a)-a)^r a^0\right]a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-a)-a)^b a^{r-b}\right] + \binom{r}{r}\right.}^{l \text{ sums}}$$

$$\left.\cdots\left[\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-la)^b a^{r-b}\right] + \binom{r}{r}(x-la)^r a^0\right]\cdots a^0\right]a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right]}^{l \text{ sums}}$$

$$+\binom{r}{r}a^0\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2a)^b a^{r-b}\right]$$

$$+\binom{r}{r}^2 a^{(0)^2}\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-3a)^b a^{r-b}\right]$$

$$+\cdots+\binom{r}{r}^{l-1} a^{(0)^{l-1}}\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-la)^b a^{r-b}\right]$$

$$+\binom{r}{r}^{l}(x-la)^r a^{(0)^l}$$

$$x^r = \sum_{b=0}^{l-1}\left[\binom{r}{r}^b a^{(0)^b}\sum_{c=0}^{r-1}\left[\binom{r}{c}(x-(b+1)a)^c a^{r-c}\right]\right] + \binom{r}{r}^l(x-la)^r a^{(0)^l}$$

note: for the simplifed sum aboe to work binom(r,r)^(0) and a^(0)^(0) must simplify to 1, so a^0 is implictly assumed to be 1 for that sum to work. you could factor out first term of the sum to make it pure

if we add restrictions the sum simplifies

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$$\binom{r}{r} = 1 \text{ for all } r \in \mathbb{Z}^+$$

$$\text{for all } a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + (x-la)^r$$

falling sums, free upper bounds

if we just keep exapnding the new outer term derived from the falling l,r formula with the falling l,r formula we get more nested sums

$$\text{for all } a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + (x-la)^r$$

$$x^r = \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} (x-(b+1)a)^c a^{r-c} \right] \right] + \sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} ((x-la)-(b+1)a)^c a^{r-c} \right] \right] + ((x-la)-la)^r$$

$$x^r = \sum_{d=0}^{m-1} \left[\sum_{b=0}^{l-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} ((x-dla)-(b+1)a)^c a^{r-c} \right] \right] \right] + (x-mla)^r$$

$$x^r = \sum_{b_3=0}^{n_3-1} \left[\sum_{b_2=0}^{n_2-1} \left[\sum_{b_1=0}^{n_1-1} \left[\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} (((x-b_3n_2n_1n_0)-b_2n_1n_0)-(b_1+1)n_0)^{b_0} n_0^{r-b_0} \right] \right] \right] \right] + (x-n_3n_2n_1n_0)^r$$

if we expand n times

$$x^r = \sum_{b_n=0}^{n_n-1} \left[\dots \sum_{b_4=0}^{n_4-1} \left[\sum_{b_3=0}^{n_3-1} \left[\sum_{b_2=0}^{n_2-1} \left[\sum_{b_1=0}^{n_1-1} \left[\sum_{b_0=0}^{r-1} \left[\right. \right. \right. \right. \right. \right.$$

$$\left. \left. \left. \left. \left. \left. \binom{r}{b_0} ((((\dots(x-b_nn_{n-1}\dots n_3n_2n_1n_0)\dots - b_4n_3n_2n_1n_0) - b_3n_2n_1n_0) - b_2n_1n_0) - (b_1+1)n_0)^{b_0} n_0^{r-b_0} \right] \right] \right] \right] \right] \right] + (x-n_n\dots n_4n_3n_2n_1n_0)^r$$

this can be cleaned up if we avoid $((x-a)+a)^r = x^r$ substituion

$$(x+a)^r = \sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} x^{b_0} a^{r-b_0} \right] + \binom{r}{r} a^0 x^r$$

$$x^r = -\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} x^{b_0} a^{r-b_0} \right] + (x+a)^r$$

$$x^r = -\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} x^{b_0} a^{r-b_0} \right] + \left[-\sum_{b_0=0}^{r-1} \left[(x+a)^{b_0} a^{r-b_0} \right] + ((x+a)+a)^r \right]$$

$$x^r = \sum_{b_1=0}^{n_1-1} \left[-\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} (x+b_1a)^{b_0} a^{r-b_0} \right] \right] + ((x+n_1a)^r$$

$$x^r = \sum_{b_1=0}^{n_1-1} \left[-\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} (x+b_1n_0)^{b_0} n_0^{r-b_0} \right] \right] + ((x+n_1n_0)^r$$

the restrictions we have are from $\binom{r}{r}(n_0)^0$ being applied to each new varible. factoring out the -1 we get:

for all $r \in \mathbb{Z}^+ \mid n_0 \in R \mid n_0 \neq 0 \mid n_1 ... n_n \in \mathbb{Z}^+$

$$x^r = -1 \sum_{b_n=0}^{n_n-1} \left[... \sum_{b_2=0}^{n_2-1} \left[\sum_{b_1=0}^{n_1-1} \left[\sum_{b_0=0}^{r-1} \left[\binom{r}{b_0} (((x + b_n n_n ... n_1 n_0) + ... + b_2 n_1 n_0) + b_1 n_0)^{b_0} n_0^{r-b_0} \right] \right] \right] + ((x + n_n ... n_2 n_1 n_0)^r$$

binomial recurssion

hockey stick

pascals identity: $\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

pasclas identity can be applied to its own terms

i will apply the idenitty to its left hand terms

$$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$$

$$\binom{a}{n} = \left[\binom{a-2}{n} + \binom{a-2}{n-1} \right] + \binom{a-1}{n-1}$$

$$\binom{a}{n} = \left[\left[... \left[\binom{a-a}{n} + \binom{a-a}{n-1} \right] ... + \binom{a-3}{n-1} \right] + \binom{a-2}{n-1} \right] + \binom{a-1}{n-1}$$

$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\binom{b}{n-1} \right] + \binom{0}{n}$$

the hockey stick identity: binom_hockey

hockey stick recurssion

the hockey stick identity can also be applied to its own terms o get a nested sum with falling uper bounds

$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\binom{b}{n-1} \right] + \binom{0}{n}$$

$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\binom{c}{n-2} \right] + \binom{0}{n-1} \right] + \binom{0}{n}$$

$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[... \sum_{n_n=0}^{n_{n-1}-1} \left[\binom{n_n}{n-n} \right] + \binom{0}{n-(n-1)} ... \right] + \binom{0}{n-1} \right] + \binom{0}{n}$$

if we restrict n the sum simplifies

this simplifes the $\binom{0}{n-d}$ at each layer

for all $n \in \mathbb{Z}_{\geq 0}$

$$\binom{0}{n} = \begin{cases} 1 \text{ if } n = 0 \\ 0 \text{ if } n \neq 0 \end{cases}$$

this simplifes the innermost $\binom{n_n}{0}$

for all $a \in \mathbb{R}$

$$\binom{a}{0} = 1$$

when $n \neq 0$ the inner $\binom{0}{n-d}$ simplifies to 0, when n=0, the inner sums are never reached and $\binom{a}{0} = \binom{0}{0}$ hockey stick recurssion identity:

for all $n \in \mathbb{Z}_{\geq} 0$

$$\binom{a}{n} = \overbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\ldots\sum_{n_n=0}^{n_{n-1}-1}[1]\ldots\right]\right]}^{n \text{ sums}} + \binom{0}{n}$$